

Task 4: Password Security & Authentication Analysis

Tools:

- Primary: Hashcat, John the Ripper
- Alternatives: Online Hash Identifiers

Hints / Mini Guide:

1. Learn how passwords are stored (hashing vs encryption).
2. Identify different hash types (MD5, SHA-1, bcrypt).
3. Generate password hashes.
4. Attempt cracking weak hashes using wordlists.
5. Understand brute force vs dictionary attacks.
6. Analyze why weak passwords fail.
7. Study MFA and its importance.
8. Write recommendations for strong authentication.

Deliverables:

- Password security analysis report

Final Outcome:

- Knowledge of password attacks and defenses

Interview Questions Related To Above Task:

- What is hashing?
- Difference between hashing and encryption?
- What is brute force attack?
- Why is MFA important?
- What makes a strong password?

📌 Task Submission Guidelines

- 🕒 **Time Window:**

You can complete the task anytime between 10:00 AM to 10:00 PM on the given day. Submission link closes at 10:00 PM

- 🔍 **Self-Research Allowed:**

You are free to explore, Google, or refer to tutorials to understand concepts and complete the task effectively.

- 🔧 **Debug Yourself:**

Try to resolve all errors by yourself. This helps you learn problem-solving and ensures you don't face the same issues in future tasks.

- 💰 **No Paid Tools:**

If the task involves any paid software/tools, do not purchase anything. Just learn the process or find free alternatives.

- 📁 **GitHub Submission:**

Create a new GitHub repository for each task.

Add everything you used for the task — code, datasets, screenshots (if any), and a short README.md explaining what you did.

- 📤 **Submit Here:**

After completing the task, paste your GitHub repo link and submit it using the link below:

- 👉 [[Submission Link](#)]

Best
of
Luck

